

Powerful numbers between consecutive squares

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Abstract

Let $h(n)$ be the number of powerful integers in $(n^2, (n+1)^2)$. We record two complementary bounds. On the lower side we prove a *frequency* (counting) bound, uniform as $V \rightarrow \infty$: there are effective constants $c, C > 0$ with $\#\{n \leq x : h(n) \geq V\} \geq x \exp(-CV \log(2V) \log \log(3V))$ whenever $V^2 \log(2V) \log \log(3V) \leq c \log x$, so in particular $\#\{n \leq x : h(n) \geq V\} \geq x^{1-o(1)}$ for $2 \leq V \leq (\log x)^{1/2-\varepsilon}$. On the upper side we record $h(n) \ll_{\varepsilon} n^{6/25+\varepsilon}$, improving the elementary $n^{2/5}$ and the recorded unconditional $O(n/\log n)$. The construction underlying the lower bound is that of De Koninck–Luca–Shparlinski with the standard primorial modulus and is not claimed new; the upper bound is an application of the classical integer-points-close-to-a-curve method of Swinnerton-Dyer and Filaseta–Trifonov, whose only novelty is the explicit worst-case ($\theta = 0$) statement. The new mathematical point is the reduction of the frequency question to simultaneous equidistribution of the multiquadratic directions $\{1/\sqrt{d}\}$, made effective by a Liouville bound. The elementary and algebraic core of the lower bound is formalized in Lean 4/Mathlib. Both sides leave a gap to the conjectured $(\log n)^{1+o(1)}$, governed in each case by an equidistribution input beyond current technology.

1 Introduction

A positive integer is *powerful* (squarefull) if every prime dividing it does so to power at least two. Let $h(n) = \#\{\text{powerful } m : n^2 < m < (n+1)^2\}$. (Erdős’s formulation counts the left endpoint n^2 ; our open-interval normalization differs by exactly one, as n^2 is always powerful, which is immaterial for the estimates below.) Throughout, $\|x\|$ denotes the distance from $x \in \mathbb{R}$ to the nearest integer, $\|a\|_{\infty} = \max_j |a_j|$, and $e(t) = e^{2\pi it}$. Erdős asked for the order of $h(n)$ and, in the notation of the question, conjectured a constant c with $h(n) \leq (\log n)^{c+o(1)}$ for all n and $h(n) \geq (\log n)^{c-o(1)}$ infinitely often, $c = 1$ being the natural guess [1, 2].

Lower bounds, infinitely often. De Koninck–Luca [3] proved $h(n) \gg (\log n / \log \log n)^{1/3}$ for infinitely many n ; De Koninck–Luca–Shparlinski [4] extended the method to κ -full integers in $(n^{\kappa}, (n+1)^{\kappa})$, with exponent $1/3 + o(1)$; Blomer (recorded in [5, Appendix A]) improved the exponent to $1/2$. Applying Dirichlet’s theorem directly and taking the auxiliary modulus to be a primorial sharpens the infinitely-often exponent further to $1 - o(1)$, so that any admissible c above satisfies $c \geq 1$; this is a folklore optimization of the same construction (the primorial choice goes back to Ramanujan), recorded by the author at [2], which we do not claim as new and mention only for context, as it is not used below.

Distribution, fixed ℓ . Shiu [6], and Xiong–Zaharescu [7] for higher powers, proved that the density $\delta_{\ell} = \text{dens}\{n : h(n) = \ell\}$ exists for each fixed ℓ , with generating function $\sum_{\ell} \delta_{\ell} z^{\ell} = \prod_{\lambda} (1 + (z-1)/\lambda)$ over an explicit set; Narumi–Tachiya [8] treat consecutive intervals. These results are not uniform in ℓ and give no information on $\#\{n \leq x : h(n) \geq V\}$ when $V = V(x) \rightarrow \infty$.

Upper bounds. The unconditional worst-case bounds in the literature are short-interval bounds: De Koninck–Luca–Shparlinski [4] and Chan [9] give $O(y/\log y)$ powerful numbers in an interval of

length y , which at $y \asymp \sqrt{x}$ (our case, $x = n^2$) is $O(n/\log n)$; an $n^{1/2}$ -type bound holds only under the *abc* conjecture [4, Thm. 2]. Sawin observed [2] that the elementary estimate $h(n) \ll n^{2/5}$ follows from the representation of powerful numbers as a^2b^3 . The distribution of squarefull numbers in short intervals of length $x^{1/2+\theta}$ with $\theta > 0$ has been studied intensively in the *asymptotic* regime [6, 11, 13, 15], the current record being $\theta > 19/154$ [15]; but the *worst-case* count at the critical length $y \asymp \sqrt{x}$ ($\theta = 0$) does not appear to have been recorded.

We supply a frequency lower bound (Theorem 1), and record the upper bound that the standard near-curve technology yields at the critical length (Theorem 2).

Theorem 1. *There are absolute, effectively computable constants $c, C > 0$ such that the following holds. Let $H(V; x) = \#\{n \leq x : h(n) \geq V\}$. If x is large and $2 \leq V$ and $V^2 \log(2V) \log \log(3V) \leq c \log x$, then*

$$H(V; x) \geq x \exp(-C V \log(2V) \log \log(3V)).$$

Corollary 1. *For every fixed $\varepsilon > 0$, $H(V; x) \geq x^{1-o(1)}$ uniformly for $2 \leq V \leq (\log x)^{1/2-\varepsilon}$.*

Theorem 2. *For every $\varepsilon > 0$, $h(n) \ll_\varepsilon n^{6/25+\varepsilon}$.*

To the author's knowledge Theorem 1 is the first counting lower bound for $h(n)$ valid as $V \rightarrow \infty$; the existing distribution results are for fixed ℓ . Theorem 2 is not a new theorem in method: it is an application of the integer-points-close-to-a-curve technology of Swinnerton-Dyer [10] and Filaseta–Trifonov [12, 14]; we record it because the literature optimized the asymptotic ($\theta > 0$) regime and did not state the worst-case exponent, and because it answers the question of how far Sawin's elementary $n^{2/5}$ can be pushed by classical means (§9).

2 The construction and an exact one-sided window criterion

Let $D = p_1 \cdots p_h$ be the product of the first h primes and let $d_1, \dots, d_\ell$ ($\ell = 2^h - 1$) be the squarefree divisors of D exceeding 1. For $q \geq 1$ put $n = Dq$, and for each $d = d_j$ set

$$\alpha_d = d^{-1/2}, \quad r_d = \lceil q\alpha_d \rceil, \quad m_d = d(Dr_d)^2.$$

Powerfulness. Each $m_d = dD^2r_d^2$ is powerful: a prime dividing r_d occurs through r_d^2 to exponent ≥ 2 ; a prime dividing d also divides D , hence occurs in dD^2 to exponent ≥ 3 ; a prime dividing D but not d occurs in D^2 to exponent ≥ 2 . Distinct d give distinct m_d , since the squarefree kernel of m_d is exactly d .

Lemma 1. *For $d > 1$, $q/\sqrt{d}$ is irrational, so $\eta_d := r_d - q/\sqrt{d} \in (0, 1)$ and $\eta_d = 1 - \{q/\sqrt{d}\}$. With $t_d = D\sqrt{d}\eta_d$,*

$$m_d - n^2 = D^2(dr_d^2 - q^2) = 2nt_d + t_d^2,$$

and hence

$$n^2 < m_d < (n+1)^2 \iff 2nt_d + t_d^2 < 2n+1 \iff t_d < 1 \iff \left\{ \frac{q}{\sqrt{d}} \right\} > 1 - \frac{1}{D\sqrt{d}}.$$

Proof. $m_d - n^2 = D^2(q + \sqrt{d}\eta_d)^2 - D^2q^2 = 2D^2q\sqrt{d}\eta_d + D^2d\eta_d^2 = 2nt_d + t_d^2$. As $(n+1)^2 - n^2 = 2n+1$ and $t_d \geq 0$, $2nt_d + t_d^2 < 2n+1 \iff t_d < 1$; and $t_d < 1 \iff \eta_d < 1/(D\sqrt{d}) \iff \{q/\sqrt{d}\} > 1 - 1/(D\sqrt{d})$. $\square$

The ceiling $r_d = \lceil q\alpha_d \rceil$ gives a clean one-sided criterion with no tie or side ambiguity.

3 Reduction to a box-hit count

Put $\delta_d = 1/(D\sqrt{d})$, $\alpha = (d^{-1/2})_{d|D, d>1} \in [0, 1)^\ell$, and $B = \prod_{d|D, d>1} (1 - \delta_d, 1) \subset [0, 1)^\ell$. By Lemma 1, if $q\alpha \bmod 1 \in B$ then $(n^2, (n+1)^2)$ contains the ℓ distinct powerful integers m_d , so $h(Dq) \geq \ell$. Hence, for $Q = \lfloor x/D \rfloor$,

$$\#\{n \leq x : h(n) \geq \ell\} \geq \#\{q \leq Q : q\alpha \bmod 1 \in B\}.$$

Since each prime of D divides exactly half of its 2^h divisors, $\prod_{d|D, d>1} d = D^{(\ell+1)/2}$, so

$$\text{vol}(B) = \prod_{d|D, d>1} \frac{1}{D\sqrt{d}} = D^{-\ell} \left(\prod_d d \right)^{-1/2} = D^{-(5\ell+1)/4}.$$

4 A multiquadratic Liouville bound

Lemma 2. For $a \in \mathbb{Z}^\ell \setminus \{0\}$ with $H = \|a\|_\infty$, $\|a \cdot \alpha\| \geq A^{-1}H^{-\ell}$, where $A = D(3D\ell)^\ell$.

Proof. $\gamma := D(a \cdot \alpha) = \sum_d a_d(D/d)\sqrt{d}$ is an algebraic integer of $K = \mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_h})$, $[K : \mathbb{Q}] = 2^h = \ell + 1$. Let m be the nearest integer to $a \cdot \alpha$. Then $\gamma - Dm$ is a nonzero algebraic integer (nonzero because $\{1\} \cup \{\sqrt{d} : d \mid D, d > 1\}$ is $\mathbb{Q}$ -linearly independent, so $a \cdot \alpha \notin \mathbb{Z}$), whence $|\text{N}_{K/\mathbb{Q}}(\gamma - Dm)| \geq 1$. With $S = \sum_d d^{-1/2} \leq \ell/\sqrt{2}$ and $|m| \leq HS + \frac{1}{2}$, every conjugate satisfies $|\sigma(\gamma - Dm)| \leq \sum_d |a_d|(D/d)\sqrt{d} + D|m| \leq HDS + D(HS + \frac{1}{2}) \leq 3HD\ell =: M$. Separating the identity embedding from the other ℓ gives $1 \leq |\gamma - Dm| M^\ell$, so $\|a \cdot \alpha\| = |\gamma - Dm|/D \geq D^{-1}(3D\ell)^{-\ell}H^{-\ell}$. $\square$

The exponent is ℓ , not 2^ℓ : the field is generated by $h = \log_2(\ell + 1)$ square roots, so its degree is $\ell + 1$. The bound is fully effective.

5 An effective discrepancy estimate

Proposition 1. There is an absolute $C_0 > 0$ such that for every $Q \geq 3$ the star-discrepancy of $(q\alpha)_{q \leq Q}$ in $[0, 1)^\ell$ satisfies $D_Q^* \leq C_0^\ell D\ell \log(2Q) Q^{-1/(\ell+1)}$. Consequently there is an absolute $C_1 > 0$ such that if $\log Q \geq C_1 \ell^2 \log(3D)$ then $\#\{q \leq Q : q\alpha \bmod 1 \in B\} \geq \frac{1}{2} Q \text{vol}(B)$.

Proof. By the Erdős–Turán–Koksma inequality [16], for every integer $R \geq 2$,

$$D_Q^* \leq C^\ell \left(\frac{1}{R} + \sum_{0 < \|a\|_\infty \leq R} \frac{1}{r(a)} \left| \frac{1}{Q} \sum_{q \leq Q} e(qa \cdot \alpha) \right| \right), \quad r(a) = \prod_j \max(1, |a_j|),$$

with C absolute. The geometric sum gives $|\sum_{q \leq Q} e(qa \cdot \alpha)| \leq 1/(2\|a \cdot \alpha\|)$, and by Lemma 2 $\|a \cdot \alpha\|^{-1} \leq D(3D\ell)^\ell R^\ell$ for $\|a\|_\infty \leq R$. Also $\sum_{\|a\|_\infty \leq R} r(a)^{-1} \leq (1 + 2\sum_{m \leq R} m^{-1})^\ell \leq (C \log 2R)^\ell$. Hence $D_Q^* \leq C^\ell (R^{-1} + D(3D\ell)^\ell R^\ell (\log 2R)^\ell / Q)$. Take $R = \lfloor (Q/[D(3D\ell)^\ell (\log 2Q)^\ell])^{1/(\ell+1)} \rfloor$ when this is ≥ 2 (otherwise $D_Q^* \leq 1$ trivially, absorbed by enlarging C_0). Both terms are then $\ll C^\ell (D(3D\ell)^\ell (\log 2Q)^\ell / Q)^{1/(\ell+1)}$, and since $(D(3D\ell)^\ell)^{1/(\ell+1)} \leq 3D\ell$, $D_Q^* \leq C_0^\ell D\ell \log(2Q) Q^{-1/(\ell+1)}$. The box $B = \prod_j (1 - \delta_j, 1)$ is anchored at a corner, so by inclusion–exclusion the counting error is $\leq 2^\ell Q D_Q^*$, giving $\#\{q \leq Q : q\alpha \in B\} = Q \text{vol}(B) + O(2^\ell Q D_Q^*)$. This error is below $\frac{1}{2} Q \text{vol}(B) = \frac{1}{2} Q D^{-(5\ell+1)/4}$ once $2^\ell C_0^\ell D\ell \log(2Q) Q^{-1/(\ell+1)} \leq \frac{1}{2} D^{-(5\ell+1)/4}$, i.e. (taking logarithms) once $L/(\ell + 1) \geq \frac{5\ell+5}{4} \log D + O(\ell \log \ell) + O(\log L)$, where $L = \log Q$. Since $L/(\ell + 1) - O(\log L)$ is increasing in L once $L \gg \ell$, it suffices to verify this at $L = C_1 \ell^2 \log(3D)$, which holds for an absolute C_1 . $\square$

6 Proof of Theorem 1

Choose h minimal with $2^h - 1 \geq V$ and set $\ell = 2^h - 1$, $D = p_1 \cdots p_h$, so $V \leq \ell \leq 2V + 1$. By standard explicit (Chebyshev-type) prime estimates $\log D = \theta(p_h) \ll h \log(2h) \ll \log(2V) \log \log(3V)$, hence $\ell^2 \log(3D) \ll V^2 \log(2V) \log \log(3V)$. The hypothesis $V^2 \log(2V) \log \log(3V) \leq c \log x$ gives, for c small, $\log(x/D) \geq C_1 \ell^2 \log(3D)$. Applying Proposition 1 with $Q = \lfloor x/D \rfloor \geq x/(2D)$,

$$H(V; x) \geq \#\{n \leq x : h(n) \geq \ell\} \geq \frac{1}{2} Q \operatorname{vol}(B) \geq \frac{x}{4} D^{-(5\ell+5)/4}.$$

Since $\ell \log D \ll V \log(2V) \log \log(3V)$, this is $\geq x \exp(-C V \log(2V) \log \log(3V))$. Corollary 1 follows because $V \log(2V) \log \log(3V) = o(\log x)$ for $V \leq (\log x)^{1/2-\varepsilon}$. $\square$

7 A conditional near-ceiling form

Theorem 3. *Let $D = p_1 \cdots p_h$, $\ell = 2^h - 1$, $Q = \lfloor x/D \rfloor$, and assume the simultaneous-equidistribution hypothesis $D_Q^*(\alpha) \leq 2^{-\ell-1} \operatorname{vol}(B)$. Then $\#\{n \leq x : h(n) \geq \ell\} \geq \frac{1}{2} \lfloor x/D \rfloor D^{-(5\ell+1)/4}$, which for $x \geq 2D$ is*

$$\#\{n \leq x : h(n) \geq \ell\} \geq \frac{1}{4} x D^{-(5\ell+5)/4} = x^{1 - \frac{5\ell+5}{4} \frac{\log D}{\log x} + o(1)}.$$

This is a positive power of x while $\frac{5\ell}{4} \log D < (1 - o(1)) \log x$ (below the construction's height ceiling $\ell^(x) \asymp \log x / (\log \log x \log \log \log x)$); it is $x^{1-o(1)}$ only for $\ell \log D = o(\log x)$, and at $\ell = \ell^*$ it degenerates to the infinitely-often statement.*

Proof. Under the hypothesis the corner-box counting error is $2^\ell Q D_Q^* \leq \frac{1}{2} Q \operatorname{vol}(B)$, so $\#\{q \leq Q : q\alpha \in B\} \geq \frac{1}{2} Q \operatorname{vol}(B) = \frac{1}{2} \lfloor x/D \rfloor D^{-(5\ell+1)/4}$, exactly as in the second half of Proposition 1; the displayed forms then follow as in §5–§6. $\square$

8 The upper bound

We deduce Theorem 2 from the integer-points-close-to-a-curve method. Every powerful m has a unique representation $m = a^2 b^3$ with b squarefree. The method we use is the following.

Lemma 3 (Filaseta–Trifonov [12, Thm. 4.1]). *Let $k \geq 2$ be an integer and $s \in \mathbb{Q} \setminus \{-(k-1), \dots, k-1\}$. Let $f(u) = X/u^s$ with X real independent of u , and suppose $N^s \leq X$ and $\delta \leq c_k N^{-(k-1)}$ for a sufficiently small $c_k = c_k(k, s) > 0$. Then*

$$\#\{u \in \mathbb{Z} \cap (N, 2N] : \|f(u)\| < \delta\} \ll_{k,s} X^{1/(2k+1)} N^{(k-s)/(2k+1)} + \delta X^{1/(6k+3)} N^{(6k^2+2k-s-1)/(6k+3)}.$$

Proof of Theorem 2. Write $m = a^2 b^3$ with b squarefree. If $n^2 < m < (n+1)^2$ then $a^2 b^3 < (n+1)^2$, so $a < n+1$, $b < (n+1)^{2/3}$, and $\min(a, b) \leq (n+1)^{2/5}$ (else $a, b > (n+1)^{2/5}$ give $a^2 b^3 > (n+1)^2$). We bound the contributions of $b \leq (n+1)^{2/5}$ and of $a \leq (n+1)^{2/5}$ separately; double counting only helps an upper bound, and the $O(1)$ terms with small parameter ≤ 2 are discarded. Dropping the squarefree restriction on b likewise only enlarges the count.

The b -aspect. For fixed $b \geq 2$, the condition $n^2 < a^2 b^3 < (n+1)^2$ places a in an interval $(nb^{-3/2}, (n+1)b^{-3/2})$ of length $b^{-3/2} < 1$; so there is at most one admissible a , and its existence forces $\|nb^{-3/2}\| < b^{-3/2}$. Thus the $b \leq (n+1)^{2/5}$ contribution is $\leq \#\{b \leq (n+1)^{2/5} : \|nb^{-3/2}\| < b^{-3/2}\}$. Cover $b \in (2, (n+1)^{2/5}]$ by dyadic blocks $b \in (B, 2B]$, $B = n^\beta$, $0 < \beta \leq \frac{2}{5} + o(1)$, and apply Lemma 3 with $k = 2$, $s = 3/2$, $X = n$, $N = B$, $\delta \asymp B^{-3/2}$. The hypotheses hold:

$s = 3/2 \notin \{-1, 0, 1\}$; $N^s = B^{3/2} \leq n$ since $B \leq n^{2/3}$; and $\delta \asymp B^{-3/2} \leq c_2 B^{-1}$ for $B \geq c_2^{-2}$ (boundedly many smaller B contribute $O(1)$). With $6k^2 + 2k - s - 1 = 51/2$, the bound is

$$\ll n^{1/5} B^{1/10} + B^{-3/2} n^{1/15} B^{51/30} = n^{1/5+\beta/10} + n^{1/15+\beta/5},$$

the first term dominating for $\beta \leq 2/3$. Together with the trivial block bound $\#\{b \in (B, 2B]\} \ll B = n^\beta$, the per-block count is $\ll \min(n^\beta, n^{1/5+\beta/10})$, which on $0 < \beta \leq \frac{2}{5} + o(1)$ is maximized at $\beta = \frac{2}{5}$, giving $n^{6/25+o(1)}$.

The a -aspect. For fixed a , b lies in $((n/a)^{2/3}, ((n+1)/a)^{2/3})$, an interval of length $\asymp a^{-2/3} n^{-1/3} < 1$, so at most one admissible b , requiring $\|(n/a)^{2/3}\| \ll a^{-2/3} n^{-1/3}$. Cover $a \in (2, (n+1)^{2/5}]$ dyadically, $a \in (A, 2A]$, $A = n^\alpha$, $0 < \alpha \leq \frac{2}{5} + o(1)$, and apply Lemma 3 with $k = 2$, $s = 2/3$, $X = n^{2/3}$, $N = A$, $\delta \asymp A^{-2/3} n^{-1/3}$. Validity: $N^s = A^{2/3} \leq n^{2/3} = X$ since $A \leq n$; and $\delta A/c_2 \asymp A^{1/3} n^{-1/3} \ll n^{-1/5}$ since $A \leq (n+1)^{2/5}$, so $\delta \leq c_2 A^{-1}$ for n large (boundedly many small n are absorbed). The leading term is $X^{1/5} N^{(k-s)/5} = n^{2/15} A^{4/15} = n^{2/15+4\alpha/15}$, maximized at $\alpha = \frac{2}{5}$, giving $n^{2/15+8/75} = n^{6/25+o(1)}$ (the second term is smaller).

Summing over the $O(\log n)$ dyadic blocks in each aspect costs a factor $n^{o(1)}$. Hence $h(n) \ll_\varepsilon n^{6/25+\varepsilon}$. $\square$

9 The floor of the method, and the gap

The exponent $6/25$ is attained at the symmetric scale $a \asymp b \asymp n^{2/5}$, where $m = a^2 b^3$ has both factors at the critical size. There, in the normalization of Lemma 3, the relevant window has half-width $\asymp n^{-3/5}$ and the curve has $|f''| \asymp n^{-2/5}$, so $6/25$ coincides with the on-curve lattice-point bound $N^{3/5}$ of Swinnerton-Dyer [10] at $N = n^{2/5}$.

Remark 1. For the Filaseta–Trifonov/Swinnerton-Dyer route used here, $6/25$ is the best attainable exponent. A smaller exponent would require either the higher-order ($k \geq 3$) cases of Lemma 3, which are inadmissible at the symmetric scale because the window half-width $\asymp n^{-3/5}$ exceeds the admissibility threshold $c_k N^{-(k-1)} = c_3 n^{-4/5}$ there; or an estimate of Bombieri–Pila on-curve quality ($\ll N^{1/2+\varepsilon}$, which would give $n^{1/5}$) valid for a positive window half-width in the regime where the two curve scales coincide. Trifonov [14, 15] identifies exactly this — extending Swinnerton-Dyer’s on-curve count to the positive-width case at the symmetric scale $a \asymp b \asymp n^{2/5}$ (both factors of size $\asymp X^{1/5}$ for squarefull numbers near $X = n^2$) — as the open obstacle, and no such extension is known. Thus $6/25$ is the natural limit of present technology for the worst-case count, consistent with Sawin’s expectation [2] that arguments of this kind yield a power of n rather than a power of $\log n$.

Remark 2. Numerically $\max_{n \leq 10^7} h(n) = 9$, attained at $n = 524827$, and the growth of $\max_{n \leq N} h(n)$ is consistent with $(\log N)^{1+o(1)}$. The gap between this conjectured truth and our $n^{6/25}$ on the upper side, and between it and the height ceiling $\ell^*(x)$ on the lower side (Theorem 3), is in each case governed by an equidistribution input beyond current technology: a positive-width on-curve count on the upper side, and a lower bound for $\|\sum_d a_d \sqrt{d}\|$ uniform in ℓ on the lower side. These appear related to the arithmetic of the conjectured bound $h(n) \leq (\log n)^{1+o(1)}$, though that bound could fail for other reasons.

Remark 3. The lower-bound construction is a quantitative refinement of [3, 4] with the standard primordial modulus; the new point is not the construction but the extraction of a counting lower bound for $\#\{n \leq x : h(n) \geq V\}$ uniform as $V \rightarrow \infty$, via an exact reduction to a box-hit count

(§2–§3), a discrepancy criterion (Proposition 1, second half), and the quantitative Erdős–Turán–Koksma input made effective by Lemma 2. The upper bound (§8) uses only classical near-curve technology and is recorded for the worst-case length, where the literature is silent.

Remark 4. The elementary core of the lower bound — powerfulness of the m_d , the exact window criterion Lemma 1, distinctness, and the deduction $h(Dq) \geq \ell$ from a box hit — is formalized in Lean 4/Mathlib with no `sorry` on the standard axioms `{propext, Classical.choice, Quot.sound}`, as is the abstract Liouville norm mechanism. The multiquadratic Liouville inequality (Lemma 2) and the quantitative Erdős–Turán–Koksma input (Proposition 1) are each recorded as one explicit named axiom, since neither is presently in Mathlib (both are classical, provable mathematics). Thus the formalization verifies the algebraic and elementary core and the final counting deduction modulo two stated classical inputs. The Lean development is available in the accompanying repository, built against Mathlib v4.30.0.

Use of AI

Claude (Opus 4.8), Codex (GPT-5.5) and Aristotle were used in this work, including for adversarial checking of the arguments and for the Lean formalization; the author has checked the mathematics and the cited literature and takes responsibility for all claims. The Lean kernel checks the elementary and algebraic core of the lower bound on the standard Mathlib axioms; the formal counting theorem is checked relative to the two explicit named axioms recorded above.

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